Tugas 1
McCulloct Pitts

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D4 TEKNIK KOMPUTER
POLITEKNIK ELEKTRONIKA NEGERI SURABAYA
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TUGAS TEORI

Tugas :

1. Buatlah model neuron McP untuk menyatakan fungsi logika “AND” 2 Masukan
2. Buatlah model neuron McP untuk menyatakan fungsi logika “OR” 2 Masukan
3. Buatlah model neuron McP untuk menyatakan fungsi logika “XOR” 2 Masukan
4. Buatlah model neuron McP untuk menyatakan fungsi logika “AND” 3 Masukan
5. Buatlah model neuron McP untuk menyatakan fungsi logika “OR” 3 Masukan
6. Buatlah model neuron McP untuk menyatakan fungsi logika “XOR” 3 Masukan

1. Logika AND 2 Masukan

```c
#include <stdio.h>
void main() {
    int w1=1,w2=1,t=2;
    int x1,x2,y,out;
    printf("Operasi AND\n");
    printf("Input x1=");
    scanf_s("%d", &x1);
    printf("Input x2=");
    scanf_s("%d", &x2);
    y=w1*x1+w2*x2;
    if (y >= t)
        out = 1;
    else
        out = 0;
    printf("Output=%d\n", out);
}
```

Hasil:

a. Input x1 = 0 dan x2 = 0

```
Operasi AND
Input x1=0
Input x2=0
Output=0
```

b. Input x1 = 0 dan x2 = 1

```
Operasi AND
Input x1=0
Input x2=1
Output=0
```

c. Input x1 = 1 dan x2 = 0

```
Operasi AND
Input x1=1
Input x2=0
Output=0
```

d. Input x1 = 1 dan x2 = 1

```
Operasi AND
Input x1=1
Input x2=1
Output=1
```

x1	x2	Output
0	0	0
0	1	0
1	0	0
1	1	1

2. Logika OR 2 Masukan

```c
#include <stdio.h>
void main() {
    int w1 = 2, w2 = 2, t = 2;
    int x1, x2, y, out;
    printf("Operasi OR\n");
    printf("Input x1=");
    scanf_s("%d", &x1);
    printf("Input x2=");
    scanf_s("%d", &x2);
    y = w1 * x1 + w2 * x2;
    if (y >= t)
        out = 1;
    else
        out = 0;
    printf("Output=%d\n", out);
}
```

Hasil:

x1	x2	Output
0	0	0
0	1	1
1	0	1
1	1	1

a. Input x1 = 0 dan x2 = 0

```
Operasi OR
Input x1=0
Input x2=0
Hasil=0
```

b. Input x1 = 0 dan x2 = 1

```
Operasi OR
Input x1=0
Input x2=1
Hasil=1
```

c. Input x1 = 1 dan x2 = 0

```
Operasi OR
Input x1=1
Input x2=0
Hasil=1
```

d. Input x1 = 1 dan x2 = 1

```
Operasi OR
Input x1=1
Input x2=1
Hasil=1
```

3. Logika XOR 2 Masukan

```c
#include <stdio.h>
void main() {
    int w1=2,w2=-1,w3=2,w4=-1,w5=2,w6=2,t=2;
    int x1, x2, y, o1,o2,o3,z1,z2;
    printf("Operasi XOR\n");
    printf("Input x1=");
    scanf_s("%d", &x1);
    printf("Input x2=");
    scanf_s("%d", &x2);
    o1 = w1 * x1 + w4 * x2;
    if (o1 >= t)
        z1 = 1;
    else
        z1 = 0;
    o2 = w2 * x1 + w3 * x2;
    if (o2 >= t)
        z2 = 1;
    else
        z2 = 0;
    o3 = w5 * z1 + w6 * z2;
    if (o3 >= t)
        y = 1;
    else
        y = 0;
    printf("Output=%d\n", y);
}
```

Hasil:

a. Input x1 = 0 dan x2 = 0

```
Operasi XOR
Input x1=0
Input x2=0
Output=0
```

b. Input x1 = 0 dan x2 = 1

```
Operasi XOR
Input x1=0
Input x2=1
Output=1
```

c. Input x1 = 1 dan x2 = 0

```
Operasi XOR
Input x1=1
Input x2=0
Output=1
```

d. Input x1 = 1 dan x2 = 1

```
Operasi XOR
Input x1=1
Input x2=1
Output=0
```

x1	x2	Output
0	0	0
0	1	1
1	0	1
1	1	0

4. Logika AND 3 Masukan

```c
#include <stdio.h>
int main() {
    int x1, x2, x3;
    int w1 = 1, w2 = 1, w3 = 1;
    int t = 3;
    int y, out;
    printf("Operasi AND dengan 3 Masukan\n");
    printf("Input X1 = ");
    scanf("%d", &x1);
    printf("Input X2 = ");
    scanf("%d", &x2);
    printf("Input X3 = ");
    scanf("%d", &x3);
    y = x1 * w1 + x2 * w2 + x3 * w3;
    if (y >= t)
        out = 1;
    else
        out = 0;
    printf("Output = %d\n", out);
    return 0;
}
```

Hasil:

a. Input x1 = 0, x2 = 0, x3 = 0

```
Operasi AND dengan 3 Masukan
Input X1 = 0
Input X2 = 0
Input X3 = 0
Output = 0
```

b. Input x1 = 0, x2 = 0, x3 = 1

```
Operasi AND dengan 3 Masukan
Input X1 = 0
Input X2 = 0
Input X3 = 1
Output = 0
```

c. Input x1 = 0, x2 = 1, x3 = 0

```
Operasi AND dengan 3 Masukan
Input X1 = 0
Input X2 = 1
Input X3 = 0
Output = 0
```

d. Input x1 = 0, x2 = 1, x3 = 1

```
Operasi AND dengan 3 Masukan
Input X1 = 0
Input X2 = 1
Input X3 = 1
Output = 0
```

e. Input x1 = 1, x2 = 0, x3 = 0

```
Operasi AND dengan 3 Masukan
Input X1 = 1
Input X2 = 0
Input X3 = 0
Output = 0
```

f. Input x1 = 1, x2 = 0, x3 = 1

```
Operasi AND dengan 3 Masukan
Input X1 = 1
Input X2 = 0
Input X3 = 1
Output = 0
```

g. Input x1 = 1, x2 = 1, x3 = 0

```
Operasi AND dengan 3 Masukan
Input X1 = 1
Input X2 = 1
Input X3 = 0
Output = 0
```

h. Input x1 = 1, x2 = 1, x3 = 1

```
Operasi AND dengan 3 Masukan
Input X1 = 1
Input X2 = 1
Input X3 = 1
Output = 1
```

x1	x2	x3	Output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

5. Logika OR 3 Masukan

```c
#include <stdio.h>
int main() {
    int x1, x2, x3;
    int w1 = 2, w2 = 2, w3 = 2;
    int t = 2;
    int y, out;
    printf("Operasi OR dengan 3 Masukan\n");
    printf("Input X1 = ");
    scanf("%d", &x1);
    printf("Input X2 = ");
    scanf("%d", &x2);
    printf("Input X3 = ");
    scanf("%d", &x3);
    y = x1 * w1 + x2 * w2 + x3 * w3;
    if (y >= t)
        out = 1;
    else
        out = 0;
    printf("Output = %d\n", out);
    return 0;
}
```

Hasil:

a. Input x1 = 0, x2 = 0, x3 = 0

```
Operasi OR dengan 3 Masukan
Input X1 = 0
Input X2 = 0
Input X3 = 0
Output = 0
```

b. Input x1 = 0, x2 = 0, x3 = 1

```
Operasi OR dengan 3 Masukan
Input X1 = 0
Input X2 = 0
Input X3 = 1
Output = 1
```

c. Input x1 = 0, x2 = 1, x3 = 0

```
Operasi OR dengan 3 Masukan
Input X1 = 0
Input X2 = 1
Input X3 = 0
Output = 1
```

d. Input x1 = 0, x2 = 1, x3 = 1

```
Operasi OR dengan 3 Masukan
Input X1 = 0
Input X2 = 1
Input X3 = 1
Output = 1
```

x1	x2	x3	Output
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

e. Input x1 = 1, x2 = 0, x3 = 0

```
Operasi OR dengan 3 Masukan
Input X1 = 1
Input X2 = 0
Input X3 = 0
Output = 1
```

f. Input x1 = 1, x2 = 0, x3 = 1

```
Operasi OR dengan 3 Masukan
Input X1 = 1
Input X2 = 0
Input X3 = 1
Output = 1
```

g. Input x1 = 1, x2 = 1, x3 = 0

```
Operasi OR dengan 3 Masukan
Input X1 = 1
Input X2 = 1
Input X3 = 0
Output = 1
```

h. Input x1 = 1, x2 = 1, x3 = 1

```
Operasi OR dengan 3 Masukan
Input X1 = 1
Input X2 = 1
Input X3 = 1
Output = 1
```

6. Logika XOR 3 Masukan

```c
#include <stdio.h>
int xor_gate(int x1, int x2) {
    int w1 = 2, w2 = -1, w3 = 2, w4 = -1, w5 = 2, w6 = 2, t = 2;
    int y, o1, o2, o3, z1, z2;
    o1 = w1 * x1 + w4 * x2;
    z1 = (o1 >= t) ? 1 : 0;
    o2 = w2 * x1 + w3 * x2;
    z2 = (o2 >= t) ? 1 : 0;
    o3 = w5 * z1 + w6 * z2;
    y = (o3 >= t) ? 1 : 0;
    return y;
}
int main() {
    int x1, x2, x3, _temp;
    printf("Operasi XOR 3 Masukan\n");
    printf("Input x1 = ");
    scanf("%d", &x1);
    printf("Input x2 = ");
    scanf("%d", &x2);
    printf("Input x3 = ");
    scanf("%d", &x3);
    _temp = xor_gate(x1, x2);
    printf("Output = %d\n", xor_gate(_temp, x3));
    return 0;
}
```

Hasil:

a. Input x1 = 0, x2 = 0, x3 = 0

```
Operasi XOR 3 Masukan
Input x1 = 0
Input x2 = 0
Input x3 = 0
Output = 0
```

b. Input x1 = 0, x2 = 0, x3 = 1

```
Operasi XOR 3 Masukan
Input x1 = 0
Input x2 = 0
Input x3 = 1
Output = 1
```

c. Input x1 = 0, x2 = 1, x3 = 0

```
Operasi XOR 3 Masukan
Input x1 = 0
Input x2 = 1
Input x3 = 0
Output = 1
```

d. Input x1 = 0, x2 = 1, x3 = 1

```
Operasi XOR 3 Masukan
Input x1 = 0
Input x2 = 1
Input x3 = 1
Output = 0
```

e. Input x1 = 1, x2 = 0, x3 = 0

```
Operasi XOR 3 Masukan
Input x1 = 1
Input x2 = 0
Input x3 = 0
Output = 1
```

f. Input x1 = 1, x2 = 0, x3 = 1

```
Operasi XOR 3 Masukan
Input x1 = 1
Input x2 = 0
Input x3 = 1
Output = 0
```

g. Input x1 = 1, x2 = 1, x3 = 0

```
Operasi XOR 3 Masukan
Input x1 = 1
Input x2 = 1
Input x3 = 0
Output = 0
```

h. Input x1 = 1, x2 = 1, x3 = 1

```
Operasi XOR 3 Masukan
Input x1 = 1
Input x2 = 1
Input x3 = 1
Output = 1
```

x1	x2	x3	Output
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1